

HW 3

1. Let $v, w \in \mathbb{R}^n$ be nonzero vectors

$$a) \quad v w^T = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \begin{bmatrix} w_1 & w_2 & \dots & w_n \end{bmatrix} = \begin{bmatrix} v_1 w_1 & v_1 w_2 & \dots & v_1 w_n \\ v_2 w_1 & v_2 w_2 & \dots & v_2 w_n \\ \vdots & \vdots & \ddots & \vdots \\ v_n w_1 & v_n w_2 & \dots & v_n w_n \end{bmatrix}_{n \times n} = \begin{bmatrix} w_1 v & w_2 v & \dots & w_n v \end{bmatrix}_{n \times n}$$

scalars

If we multiply a vector v with w^T , we get an $n \times n$ matrix where every column is a scalar multiple of v where $w_1, w_2, \dots, w_n$ are scalars. Thus, the rank of this matrix is one since the column space is 1-dimensional.

b) A, B are square matrices

$$A \times B = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & \dots & B_{1n} \\ B_{21} & B_{22} & \dots & B_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n1} & B_{n2} & \dots & B_{nn} \end{bmatrix} = \begin{bmatrix} AB_1 & AB_2 & \dots & AB_n \end{bmatrix}$$

let e_j be the j -th standard basis vector, where each column of B can be written as

$$B_j e_j^T \quad (\text{which is rank 1})$$

$$B = \sum_{j=1}^n B_j e_j^T$$

$$AB = A \left(\sum_{j=1}^n B_j e_j^T \right) = \sum_{j=1}^n AB_j e_j^T$$

where AB is represented as a sum of rank 1 matrices up to n

$$2. A, B \in \mathbb{R}^{n \times n}$$

$$C = \begin{pmatrix} A \\ B \end{pmatrix}$$

$$\text{Null} : Cx = 0 \Rightarrow \begin{bmatrix} A \\ B \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} Ax \\ Bx \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\hookrightarrow$ if $Ax = 0$ and $Bx = 0$, then $x \in \text{Null}(C)$

$$\therefore \boxed{\text{Null}(C) = \text{Null}(A) \cap \text{Null}(B)}$$

$$3. x \in \mathbb{C}^n : \|x\|_2 \leq \sqrt{n} \|x\|_\infty$$

We know that $|x_i| \leq \|x\|_\infty$ since $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$ for $1 \leq i \leq n$

$$\Rightarrow |x_i|^2 \leq \|x\|_\infty^2$$

Apply this inequality to all n components of x ,

$$\Rightarrow \sum_{i=1}^n |x_i|^2 \leq \sum_{i=1}^n \|x\|_\infty^2 \Rightarrow \sqrt{\|x\|_2^2} \leq \sqrt{n \|x\|_\infty^2} \Rightarrow \|x\|_2 \leq \sqrt{n} \|x\|_\infty$$

definition of $\|x\|_2^2$ sum is a constant

our original assumption $\square$

4. LU decomposition Gaussian Elimination

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix} \xrightarrow{R_3 - 3R_1} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \cdot U = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}}_L \cdot \underbrace{\begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix}}_U$$

S.o) Let $s_i = x_i y_i + \dots + x_i y_i = \sum_{k=1}^i x_k y_k$

recursive: $\hat{s}_i = f(\hat{s}_{i-1} + \hat{z}_i)$

$$\hat{z}_i = f(x_i y_i)$$

$$e_i = \hat{s}_i - s_i$$

$$f(a \circ b) = \frac{a \circ b}{1 + \delta}, \quad |\delta| \leq \epsilon_{mach}$$

$$\Rightarrow a \circ b = f(a \circ b)(1 + \delta)$$

$$\hat{z}_i = f(x_i y_i) = \frac{x_i y_i}{1 + \delta} \Rightarrow x_i y_i = \hat{z}_i (1 + \delta_i)$$

$$\hat{z}_i - x_i y_i = -\hat{z}_i \delta_i$$

$$\hookrightarrow \hat{s}_i = f(\hat{s}_{i-1} + \hat{z}_i) = \frac{\hat{s}_{i-1} + \hat{z}_i}{1 + \theta_i}, \quad |\theta_i| \leq \epsilon_{mach}$$

$$\hat{s}_{i-1} + \hat{z}_i = \hat{s}_i (1 + \theta_i) = \hat{s}_i + \hat{s}_i \theta_i$$

Define error e_i :

$$e_i = \hat{s}_i - s_i = \hat{s}_i - (s_{i-1} + x_i y_i) = \hat{s}_i - \hat{s}_{i-1} - \hat{z}_i + \hat{s}_{i-1} - s_{i-1} + \hat{z}_i - x_i y_i$$

$$= \underbrace{(\hat{s}_i - \hat{s}_{i-1} - \hat{z}_i)}_{-\hat{s}_i \theta_i} + \underbrace{(\hat{z}_i - x_i y_i)}_{-\hat{z}_i \delta_i}$$

$$= e_{i-1} + \hat{s}_i \theta_i - \hat{z}_i \delta_i$$

$$\Rightarrow |e_i| \leq |e_{i-1}| + |\hat{s}_i \theta_i| + |\hat{z}_i \delta_i| \leftarrow \text{triangle inequality}$$

$$\Rightarrow |e_i| \leq |e_{i-1}| + \epsilon_{mach} |\hat{s}_i| + \epsilon_{mach} |\hat{z}_i| \quad \text{since } |\theta_i| \leq \epsilon_{mach}, \quad |\delta_i| \leq \epsilon_{mach}$$

$$\Rightarrow |e_i| \leq |e_{i-1}| + \epsilon_{mach} (|\hat{s}_i| + |\hat{z}_i|)$$

b) # of flops;

inner product

- 1 multiplication
- 1 addition
- n iterations

$\rightarrow 2n$ flops

error bound

- 2 absolute value ops
- 2 addition
- 1 multiplication

$\rightarrow 3n$ flops

This increases the flop count by 1.5 times the amount

```

1 import numpy as np
2
3
4 def inner_product_with_error(x, y):
5     n = len(x)
6     eps_mch = np.finfo(float).eps
7
8     s_hat = 0.0
9     e_bound = 0.0
10
11     for i in range(n):
12         # compute floating point product z_i
13         z_hat = x[i] * y[i]
14
15         # update partial sum s_i
16         s_next = s_hat + z_hat
17
18         # update the running error bound
19         e_bound = e_bound + eps_mch * (abs(s_next) + abs(z_hat))
20
21         s_hat = s_next
22
23     return s_hat, e_bound
24
25
26 x = np.random.randn(1000)
27 y = np.random.randn(1000)
28 val, error_limit = inner_product_with_error(x, y)
29
30 print(f"Result: {val}")
31 print(f"Running Error Bound: {error_limit}")
32
33 ''' TERMINAL OUTPUT
34 Result: 3.126331331626953
35 Running Error Bound: 5.129872869599341e-12
36
37 Result: -29.611073976476906
38 Running Error Bound: 5.275048756180587e-12
39
40 Result: 44.69289291092224
41 Running Error Bound: 7.330936719092104e-12
42 '''

```